



Dr. Alessandro Agostini

Homework 1

Issued Sunday 29 September. Due as **zipped archive with PDF+TEX into eCLASS by Monday 7 October 2024 at noon (12:00, system's clock)**. Hardcopy: none (upon request only for specific cases).

Please note the following:

0. Deadline is **strict**. **Electronic deadline passed without submission implies mark 0 (zero)**. (Equivalently, you will not be admitted to enter the classroom for Quiz 1.)

1. The suggested number of students is 1 to 3. We allow teams formed by maximum 4 members.

1.1 Teams formed by more than 4 members **are not permitted** (students not admitted to Quiz 1).

2. Although teams can be formed by students attending any of the sections 001 to 004, it is preferable all students are in the same section. In any case, students are required to write their **section number** — example: 001 — in the cover of submitted work (see 3 below). NB: Do *not* use the group. Use section!

3. L^AT_EX template. Submitted homework must be set using the provided L^AT_EX's template (see (*)).

Note: You are asked to submit a **zipped archive** containing both the source L^AT_EX file (.tex file) and its compilation file (.pdf file) using command *pdflatex*.

IMPORTANT: Only students listed in the cover of submitted work are admitted to Quiz 1. In other words, students whose ID, full name and section number do not appear on the cover **are not admitted to enter the classroom for Quiz 1 and are assigned score 0/100 for the homework**.

(*) The document (pdf file) you submit must be produced by using L^AT_EX, specifically one of following versions: 1. <https://ctan.org/pkg/texlive> (general) or 2. <https://ctan.org/pkg/mactex> (MacOS users). You are asked to send a **zipped archive** (format .zip) containing both the source file (format .tex) and the equivalent PDF file (object file, resulted by execution of command *pdflatex* on the source file).

4. Only one member of the team submits the document.

IMPORTANT: Multiple submission by members of the same team **are penalized** as follows: For each team member who submits the document already submitted, scoring (of related Quiz 1) is reduced by 5/100 points for all team members. **Example:** A team of three, three members submit: score of Quiz 1 reduced by 10 points (over 100 points) for all team members.

4.1 The **name of the submitted zipped archive as well the name of each submitted file, namely, .tex file and .pdf file** is required to be in the following format:

00X-ID (example: 001-U1910123), where:

- 00X is the section of the team's member, say S, who submits the homework to eCLASS;
- ID is the ID number of S.

5. IMPORTANT: Homework that contains evident elements of **plagiarism or cheating**, either among teams or from existing material (e.g., from the Internet, textbook's Instructor Manual if available online, etc.) implies a mark 0/100 **for all teammates and all future homework assignments**. (Equivalently, you will not be allowed to enter any of in-class quizzes.)

5.1 IMPORTANT: Sharing of homework's file(s) among teams is never permitted at whatever stage of work in progress and it results in a main source of assigned zeros. **You can share ideas, not files!**

PARTICULAR SETTING:

To proceed, you need to install MYSQL SERVER and the graphical interface/console *phpMyAdmin* for administering and browsing MySQL (or any compatibly other) database server.

IMPORTANT: You are not allowed to use any other **interface** than *phpMyAdmin*. Please read the slides, ask to a classmate, or contact the Instructor or TA if you need help at this stage.

By using *phpMyAdmin*, create a database (give the name you like; below we used “University DB”) by importing the ‘DDL for MySQL’ and ‘smallRelationsInsertFile.sql’ files that you previously downloaded from textbook’s website: <https://www.db-book.com/db6/>; cf. slides for Week 2 (see Appendix) for installation guidelines. The logical schema of such database is showed in Figure 1 (cf. last page).

PROBLEMS AND EXERCISES:

1. For **each SQL query**, say Q , in the slides discussed in class, exactly from slide 51 in slide-set of Week 3 to slide 101 in slide-set of Week 4 (both slide-sets are available in eClass), do:
 - (a) Rewrite Q into your document by using **plain SQL** as written in the slides. Next to the query write the slide-set (example: W3 for slides of Week 3; W4 for slides of Week 4,...) and the slide’s page number (you find it bottom-right corner of each slide) of the slide where Q is written. Example: if Q is the SQL query written in the slide with page number 52 of Week 3, you could write something like W3-p52 next to Q .
 - (b) By using *phpMyAdmin*’s Console, rewrite Q and screen-shot it. Then embed the screen-shot in your document just below the query written in plain SQL (cf. previous point).
NOTE: You can screen-shot it together with the result of its execution, see (d) below.
 - (c) Execute the query over the database “University DB”.
 - (d) Screen-shot the result of execution (table’ schema and instance). Two cases arise:
Case 1. Execution leads to result. Then embed the screen-shot of such result in your document just below the screen-shot of the query you embedded (cf. point (b) above).
Case 2. Execution leads to “syntax error” or any other execution error. Then:
 - debug your query (refer to MYSQL’s manual, if necessary) and execute it again (and again, if necessary) until you managed to solve the error and correctly execute the query;
 - screen-shot **both the query you finally executed and its result** (table’ schema and instance); embed it into your document just below the screen-shot of the query you embedded in (b) above (i.e., the original query Q);
 - explain in short what kind of issue you had to correctly execute the query. Be brief yet precise to identify the issue(s) you faced.
2. Write **twenty queries in English** over the database represented by schema in Figure 1 (cf. last page). Each query should be **as complex as possible**, provided that it is expressible by using SQL’s clauses and statements advanced in the slides discussed in class from slide 51 in slide-set of Week 3 to slide 101 in slide-set of Week 4.

IMPORTANT: The average difficulty of the proposed twenty queries is an integral element of this homework's evaluation.

Then, for each query you wrote, do:

- (a) Write an *equivalent* SQL query (according to definition below) by using plain SQL (i.e., SQL statements and structure discussed in class and in the slides).

Definition: Two queries (in whatever language you express each of them) on a (relational) database DB are *equivalent* if they produce (represent, compute, process) exactly (a) the same structure (schema) and (b) the same content (instance) over DB' schema and data instance.

Remark: Actually, your goal here is to write a *linguistically equivalent* query, namely, an equivalent query that at best aims to capture *all linguistic elements* (syntax and semantics) of English into the SQL query. However, remember that it could be not possible to do it, due to the limited expressivity of the database schema.

Example: The term “registered” in English query iii. below can't be directly, i.e., linguistically, expressed in a SQL query, since no table in the schema of “University DB” has the name explicitly using such term or any variant of it. In such case and similar cases, the SQL query you can write will only approximate the meaning of the English query, according to the database schema available and how the programmer (i.e., who writes the SQL query: you) interpretes the English query in input.

- (b) By using the Console of *phpMyAdmin*, execute the query on the “University DB” created in the section “Particular Setting” above. If a syntax error is prompted by the system, understand the error, rewrite the query and execute it again. Iterate the process, if necessary.

Finally:

- 1 - write in plain text the SQL query **as modified to run without errors** on *phpMyAdmin*;
- 2 - embed in the document the screenshot of the query's result (table' schema and instance).

See the next page for the database's schema to use for querying.

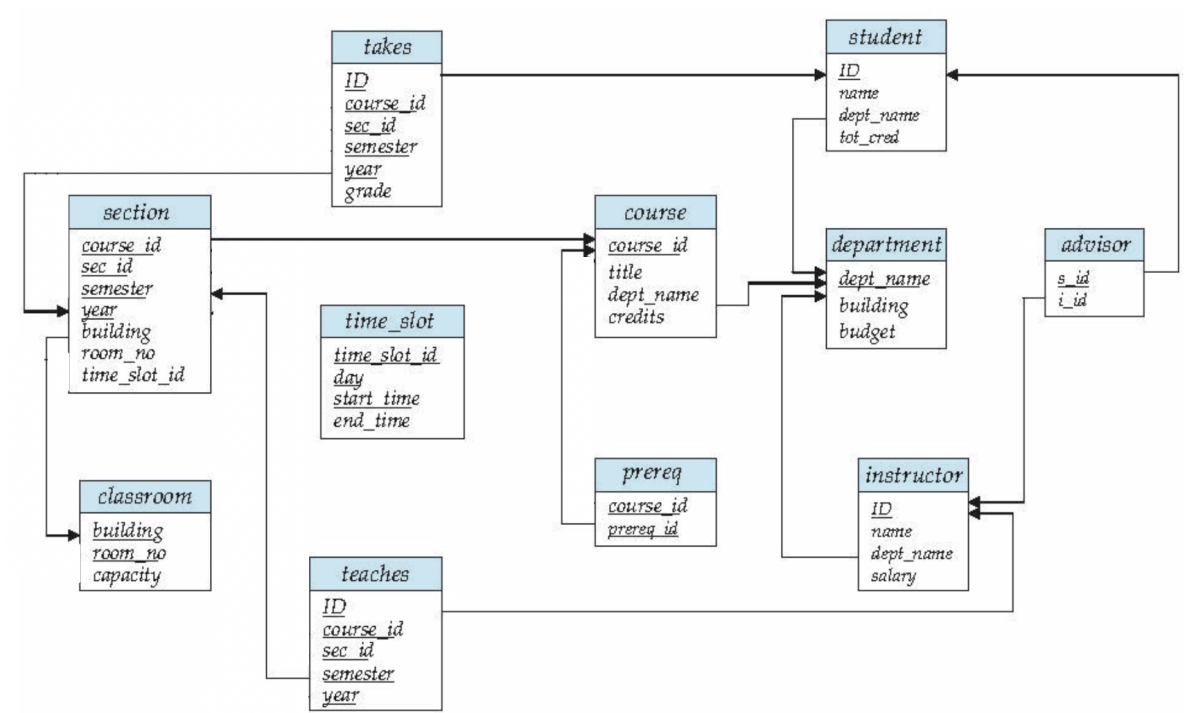


Figure 1: University Database (logical schema).

End of Homework 1